Hints Dilute and Rotate the Evidence Axis of Vision-Language Models

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Abstract

Vision-language models often answer when the image lacks the queried evidence, yet retract a correct answer as soon as a prompt sentence contradicts it. Are these failures of seeing or of reporting? We use a 2×3 within-subject design—visual evidence {present, annotation-absent} $\times$ textual hint {none, correct, conflicting}—built without editing a pixel: "absent" is a category that COCO annotations place outside the photograph, checked by a presence probe. Across 261 images and three VLMs built on Qwen2/2.5 language models, conflict collapses native signals: $p(\texttt{unknown})$ falls from AUROC 0.993 to 0.834 in Qwen2.5-VL-3B, and maximum probability reaches chance (0.463). A probe fitted only on the no-hint pair keeps its ranking on held-out scenes (0.96, 0.94, 0.92 in 3B, 7B and LLaVA, plausible hint), so the evidence is still there; what changes is its geometry. The full present–absent mean-difference vector shortens to 0.39–0.64 of its no-hint length and turns 24°–71° away from the no-hint axis, with the two hinted colours rotating to nearly the same direction ($\cos = 0.91$); per-scene gaps lose coherence ($0.70 \rightarrow 0.41$ in 7B), while noise-normalised separation does not fall and in-condition refitting recovers at least 0.989. A fourth backbone outside that family (Gemma-3-4B, Gemma-3 with a SigLIP encoder) repeats the geometry—0.53 of the length, 24°–29° of rotation, the two hinted colours at $\cos = 0.97$, 0.96 held-out transfer—but not the confidence collapse, so the geometry, not the output collapse, is the part that generalises. The hint does not erase the evidence; it moves it off the axis the readout was using, and dilutes it. Signal detection separates dilution from criterion shift: sensitivity falls in all three Qwen-family models (d' 3.51 $\rightarrow$ 1.45, 2.98 $\rightarrow$ 1.24, 3.44 $\rightarrow$ 1.44), criterion shifts are inconsistent, most of the flattened curve is the axis. Patching localises recovery to the last few layers (80% at 89% depth in 3B; 90% and 96% at 93% in 7B and LLaVA). Steering shows readout-level sufficiency; energy-matched removal shows no direction-specific necessity, so we claim an efficient readout direction, not a circuit. A frozen gate cuts conflict fabrication at 50% coverage from 0.249 to 0.069 (3B), 0.268 to 0.126 (7B) and 0.295 to 0.111 (LLaVA); conformal certification at 10% risk holds for two of three models; on naturally unanswerable VizWiz questions $p(\texttt{unknown})$ matches the probe. Code is released.

1 Introduction

Ask a vision-language model (VLM) *"What color is the umbrella?"* about a photograph with no umbrella, and it will usually answer `unknown`. Prepend one plausible sentence—*"Hint: the umbrella is black."*—and the same model answers *black* with high confidence. The failure is not that the model cannot see. The object was never there.

Two literatures describe adjacent phenomena. Work on hallucination measures objects a model reports but the image does not contain [18, 23]. Work on sycophancy measures answers that follow a user's stated belief rather than the input [12, 24, 27]. Both describe behaviour; neither separates two questions that the behaviour confounds: *is the evidence represented?* and *is the decision to abstain driven by that representation?* A model that answers *black* may have lost the evidence, or it may still represent "no umbrella" while a later stage overrides abstention.

We separate the two with a design in which the image is held fixed and only the *epistemic status* of the question changes. Each of 261 photographs appears in six cells: the queried object is present or annotated-absent, crossed with no hint, a correct hint, or a conflicting hint. The absent arm requires no image editing: it asks about a category that COCO's instance annotations place outside the image. Every cell shares the same pixels, and the six prompts differ only in what they assert. A separate yes/no probe assigns mean $p(\text{yes}) = 0.095$ to the absent images versus 0.95 to the originals; annotation absence is independently checked, but it is not identical to model-perceived absence.

The design yields a dissociation (Figure 1). With no hint, the model's own $p(\texttt{unknown})$ detects evidence availability almost perfectly (AUROC 0.993). Under conflict every output-space signal degrades ($p(\texttt{unknown})$ 0.834, maximum probability 0.463, 12-way readout 0.923), while a probe fitted only on the no-hint pair keeps its ranking on scenes it never saw (0.995 refitted, 0.958 held out, hint `b`) yet its present–absent geometry changes sharply. Measuring the full mean-difference vector, rather than its projection onto one direction, separates the ways a separation can shrink. Under a conflicting hint the vector shortens to 0.53 of its no-hint length and turns 40° away from the no-hint axis in 3B; the same rotation appears in 7B and LLaVA, and in a non-Qwen backbone (24°–29° in Gemma-3-4B), and two different hinted colours rotate to almost the same direction ($\cos = 0.91$), so it is systematic rather than colour noise. The along-axis projection, which is what a fixed readout sees, falls further than either factor alone (0.41 in 3B), while noise-normalised separation does not fall and a readout refitted inside the condition recovers 0.995–0.999. The hint dilutes and rotates the evidence axis; it does not erase it. Three tests follow. Signal detection shows a sensitivity loss with inconsistent criterion shifts, most of the flattened refusal curve being predicted by the dilution. Patching restores most of the lost abstention only in the final few layers, and a direction injected at the readout controls abstention bidirectionally; we claim sufficiency at the readout, not an identified circuit, because removal at matched energy shows no direction-specific necessity. A frozen gate trained only on the no-hint pair then cuts fabrication under conflict by a factor of two to four at half coverage, a gain that survives prevalence shift but not the move to naturally unanswerable images.

Contributions.

- **A controlled construction.** A 2×3 within-subject benchmark with no image editing, in which "evidence absent" is annotation-absent and independently checked by a presence probe, and in which the conflicting hint is plausibility-matched to the present arm. The 261-item manifest and labels are frozen and released.
- **A rank–scale dissociation with a bounded causal claim.** Native output signals collapse under conflict while the internal evidence representation stays rank-decodable on held-out scenes but dilutes and rotates. We measure the full mean-difference geometry rather than a single projection, report scene-held-out transfer rather than refitted performance as the headline, and downgrade the causal claim to readout-level sufficiency after an energy-matched removal audit.
- **A deployment result and its boundary.** A frozen linear gate reduces fabrication by 2.1–3.6× across three models at 50% coverage, survives prevalence shifts from 5% to 50%, is conformally certifiable at 10% risk in two models, and loses its advantage on naturally unanswerable images.

2 Related Work

Sycophancy and prompt-induced error. Language models shift answers toward a user's stated view [24], and their explanations can diverge from the evidence that drove the answer [27]. Recent work documents the multimodal version: VLMs change visual answers to agree with textual assertions, sometimes while still encoding the original content [12]; the phenomenon is also described as blind faith in text [8]. These studies are behavioural or input-level. We intervene on hidden states, decompose the decision into sensitivity and criterion, and use the resulting direction to build a gate.

Abstention and selective prediction. Selective prediction trades coverage for risk using a confidence score [9], and has been studied for VQA under distribution shift and on unanswerable questions [28]. ReCoVERR reduces over-abstention by acquiring more evidence [25], black-box consistency identifies unreliable answers [15], and Rep2Act trains a representation-to-action router that turns latent answerability into an explicit abstention action [11]. These methods change the decision procedure or the training objective. We hold both fixed and ask *which representation* the scalar should be computed from, and show that the model's own abstention probability is the wrong representation when a hint is present.

Distribution-free risk control. Conformal prediction and conformal risk control give finite-sample guarantees for a chosen loss under exchangeability [2, 6]. We use split conformal risk control with a binomial upper confidence bound to certify a fabrication rate, and report where the calibration set is too small to certify the requested level. BCEA couples conformal filtering with evidence acquisition [29], and A^2Safe grounds VQA safety

decisions in a multi-agent evidence board [30]; we certify a fabrication rate under prompt pollution without new evidence or an agent layer.

Probing and steering internal states. Linear probes recover truthfulness, knowledge and uncertainty from hidden states [1, 4, 7, 14, 20]. Difference-of-means directions can steer behaviour [17, 22, 26, 32], and chat-model refusal is mediated by a low-dimensional direction [3, 13]. Activation steering has been applied to VLM refusal [31]. We study a different axis—whether the evidence needed to answer is available—and pair steering with a projection audit that bounds what the ablation can show.

Localisation and hallucination. Activation patching replaces a state in one forward pass with a state from another and measures the effect on the output [21]. Object hallucination is typically measured by asking about absent objects [18, 23]. We adopt the logic of absent-object questions but make the comparison within-image and within-question, vary only the textual assertion, and add an internal readout.

3 Design and Data

3.1 The 2×3 design

Each sample is a photograph, a queried category c, and a gold colour g. The six conditions are:

	textual hint		
visual evidence	none	correct	conflicting
object present	`sufficient|none`	`sufficient|correct`	`sufficient|wrong`
object absent	`absent|none`	`absent|a`	`absent|b`

In the present arm the correct hint states g and the conflicting hint states a fixed far-hue distractor. In the absent arm there is no correct colour: `a` is the absent category's most frequent colour in COCO val (plausible for the category), and `b` applies the same far-hue mapping to that colour. `b` is therefore plausibility-matched to `sufficient|wrong`, which also uses a far hue; we use `b` for the geometry contrast and report `a` separately throughout. The plausible hint `a` is the harder deployment case (it induces more credulity) and is the one used for the gate. Every image appears in all six cells, so all comparisons are within-image and within-question.

3.2 Construction without image editing

Starting from COCO val2017 [19], we retain single annotated instances with box short side $\geq$ 44 px, area fraction 1.5–35%, and mask/box fill ≥ 0.45. Colour is proposed as the area-dominant colour inside the instance mask, with purity and margin gates, then verified by hand on contact sheets. Of 1447 geometric candidates, 413 passed the automatic colour gate and 261 survived manual verification (blue 88, white 79, black 45, green 25, red 15, yellow 9; Appendix C).

For the absent arm we sample a category with no annotation in the image (74 distinct categories appear) and use that category's {first, second} most frequent colour in val as the hint. The colour is plausible for the category; we do not constrain the category to be plausible for the scene. No pixel is edited; the same JPEG is used in all six cells.

Manipulation check. A separate yes/no question ("Is there a c in this image?") yields mean $p(\text{yes}) = 0.095$ on 260 category-matched negative images versus 0.95 on the originals. The absent arm is therefore annotation-absent and independently checked. Because 6.5% of these negatives are judged present with probability above 0.5, we report a strict-absent sensitivity analysis in Appendix D and treat "absent" as annotation-level throughout.

Absence is also not the only difference between the arms: a category that fits the scene is refused less often than one that does not. Scoring each absent category by its COCO train co-occurrence with the objects annotated in the image and splitting the items at the median, no-hint refusal is lower in the better-fitting half by 0.116 $[0.047, 0.185]$ (3B), 0.069 $[0.023, 0.115]$ (7B) and 0.132 $[0.039, 0.224]$ (LLaVA). The evidence axis is unaffected: within each half, p(`unknown`) still separates present from absent images at AUROC ≥ 0.976 and the internal probe at ≥ 0.989 (Appendix D). Scene fit moves the level of refusal, not the ordering of present against absent. The effect survives a control for category frequency, which is only weakly related to it.

3.3 Prompts, readout and signals

The system prompt asks for one word and instructs the model to answer `unknown` when the image does not contain enough information. The user prompt is

```
[Hint: the {category} is {colour}.]
What color is the {category}?
Answer with one word.
```

where the hint line is omitted in the no-hint condition. The readout is a distribution over eleven colour words plus `unknown`; all twelve candidates are single tokens in every tokenizer used. Probabilities are the softmax over their logits at the first answer position.

Output signals. We compute $p(\texttt{unknown})$, maximum probability over the twelve candidates, the top-two gap, negative entropy, and a 12-dimensional logistic readout over the full candidate distribution. These use the same forward pass as the answer.

Internal probe. A logistic regression on the final-token hidden state (2048-d for 3B, 3584-d for the two 7B models), with standardised features and $C = 0.1$. It is trained on the no-hint pair `sufficient|none` vs. `absent|none`. Cross-validation is grouped by scene (five folds) so a scene never appears in both train and test; a second analysis groups by queried category to test a category→colour shortcut. The *frozen* protocol fits on the no-hint states of the training scenes and scores the *held-out* scenes' hinted states, so no scene contributes both to fitting and to evaluation; the headline numbers use this protocol at a fixed $C = 0.1$. The *refit* protocol retrains inside each condition, still grouped by scene, and is reported only as an upper bound. The direction used for steering uses $C = 0.01$ because it is injected at a fixed norm; Appendix A records both settings and the sensitivity to C.

Steering and patching. The direction w is the logistic-regression probe direction fitted on the no-hint pair. We inject $\alpha\|h\|w$ at the final position, where $\|h\|$ is the median hidden-state norm and $\alpha \in \{\pm 0.25, \pm 0.5\}$, and ablate by projecting w out of the state. A split-half check fits w on 130 scenes and evaluates on the remaining 131. For patching, layer L receives the final-token state from a no-hint pass and the remaining layers run; a shuffled-source control pastes another sample's no-hint state at the same layer.

Metrics. Evidence detection is AUROC over the 522 paired items, with scene-grouped bootstrap confidence intervals (2000 resamples) and paired bootstrap intervals for differences. Selective answering is reported as the fabrication rate among answered items at fixed coverage; fabrication at 50% coverage is the primary operating point. We report unadjusted 95% intervals and treat them as descriptive: the primary contrasts exclude zero by wide margins, while the own-axis residual reported below is marginal and is labelled as such.

Models. Qwen2.5-VL-3B and 7B [5] in bfloat16 with SDPA attention and at most 1024^2 pixels, and LLaVA-OneVision-7B [16] with the longest image side at most 512 px, loaded through `AutoModelForImageTextToText`. All LLaVA stages use the same 512 px preprocessing. All experiments ran on a single 24 GB A10.

4 Results

4.1 Behaviour: a hint moves both arms in opposite directions

Table 1 shows the basic pattern. Without hints the models are conservative in the right way: 3B refuses 90.4% of absent-object questions and 1.1% of present-object ones. A conflicting hint moves both arms toward the assertion: refusal on the absent arm falls to 52.1% (hint a) and 56.3% (hint b), and 43–47% of answers name the hinted colour. On the present arm the same hint raises refusal from 1.1% to 8.8% and drops accuracy from 76.2% to 38.3%. A correct hint does the opposite, raising accuracy to 97.3%. The hint is therefore treated as evidence, not noise: it helps when true and hurts when false. The two absent hints differ in effect size but not in sign, and the far-hue hint `b` is the closer match to the present-arm manipulation.

The base policies differ sharply. Qwen2.5-VL-7B over-refuses under conflict (55.6% on present items) and adopts the hint on only 17.6%; LLaVA-OneVision is close to fully credulous (present refusal 0.4%, absent-hint

Table 1: Behaviour on 261 verified samples (Qwen2.5-VL-3B). "Adopts hint" is the fraction whose argmax equals the hinted colour. On absent rows, 1 − refusal is the unanswerable answer rate (UAR); on present rows, refusal is the over-refusal rate (OAR). Cross-model refusal is reported in the text.

condition	refusal	mean conf.	adopts hint	accuracy
present · correct hint	0.004	0.943	—	0.973
present · conflicting hint	0.088	0.674	0.448	0.383
present · no hint	0.011	0.836	—	0.762
absent · hint a (plausible)	0.521	0.699	0.467	—
absent · hint b (far hue)	0.563	0.707	0.433	—
absent · no hint	0.904	0.799	—	—

Table 2: Evidence detection (AUROC over 522 paired items, Qwen2.5-VL-3B) from the same forward pass. "Frozen" fits the probe on training scenes' no-hint states and scores held-out scenes' states in each condition; "refit" retrains inside the condition (still scene-grouped) and is an upper bound. For p(`unknown`) the AUROC is oriented so that higher means better evidence detection. Brackets are scene-grouped bootstrap 95% CIs.

signal	no hint	conflict (`b`)	correct
frozen final-state probe	0.995 [0.989, 0.999]	0.958 [0.939, 0.974]	0.990 [0.981, 0.997]
refit final-state probe	0.995 [0.989, 0.999]	0.995 [0.990, 0.999]	1.000 [1.000, 1.000]
p(`unknown`)	**0.993**	0.834	0.969
maximum probability	0.615	0.463 [0.413, 0.514]	0.916
12-d logit readout	0.982	0.923	0.974

adoption 76–90%). Gemma-3-4B-it—Gemma-3 with a SigLIP encoder, the only backbone here whose language model is not from the Qwen family—sits between them: it refuses 84.3% of absent items without a hint, conflict lowers this to 40.2% (hint a) and 47.5% (hint b) with adoption of 57.9% and 50.2%, and on present items the same hint is adopted on 67.4% and drops accuracy from 56.7% to 24.1%. Its colour reading without a hint is weaker than 3B's, so the present arm is noisier, but the direction of both arms is the same. All four show the dilution and rotation documented next, which is why we treat the geometry as the invariant and the behavioural form as model-specific.

4.2 Native confidence collapses; the evidence axis dilutes and rotates

Table 2 and Figure 1 contain the central result. With no hint, p(`unknown`) is already an excellent evidence detector (0.993), statistically indistinguishable from the probe (+0.002 [−0.003, +0.007]). The interesting regime is conflict. There p(`unknown`) loses 0.159 AUROC, maximum probability loses 0.152 and falls to chance, and even a learned readout over the entire output distribution loses 0.059. The frozen probe loses 0.037 and transfers at 0.958 on held-out scenes; the refit probe reaches 0.995, which shows that a decoder can still read the evidence from the conflict state but does not by itself show that the no-hint decoder does.

The distinction between frozen and refit matters for the claim. The refit number answers "is the evidence still linearly decodable?" and the answer is yes. The frozen number answers "does the direction the model would have used before the hint still rank the arms on scenes it has never seen?" and the answer is yes, with a loss that is small in AUROC but large in geometry (Section 4.3). We report the frozen number in the abstract and use the refit number only as an upper bound. Scene overlap does inflate the number, but not by much: scoring the same probe on its own fitting scenes gives 0.971 against 0.958 held out for this hint, and the gap is 0.002–0.035 across models and hints. A C grid from 3×10^{-4} to 0.1 moves the held-out value by at most 0.020 (Appendix B). The held-out value for this hint is 0.946 in 7B and 0.935 in LLaVA; the plausible hint `a` gives 0.957/0.939/0.923. Under a correct hint no signal collapses: the frozen probe reaches 0.990 and p(`unknown`) 0.969, ruling out an explanation in which the probe merely responds to the presence of a hint string.

The invariance is a statement about rank, not about the state. In probe-score units the present-minus-absent gap shrinks from 15.9 to 7.7 for the plausible hint and to 7.3 for the far-hue hint, contractions to 0.485 $[0.460, 0.511]$ and 0.463 $[0.438, 0.488]$ respectively. Within-arm standard deviations are almost unchanged (2.8 and 3.1, then 2.6 and 2.9), so in pooled-SD units the separation falls from 5.1 to 2.6. The contraction replicates in 7B (0.416 $[0.386, 0.448]$ and 0.424 $[0.393, 0.457]$) and LLaVA (0.361 $[0.337, 0.384]$ and 0.344 $[0.324, 0.365]$). This is a projection onto one direction, so by itself it cannot say whether the states moved together, shrank, or turned away from the probe. The next subsection measures the full vector and finds all three.

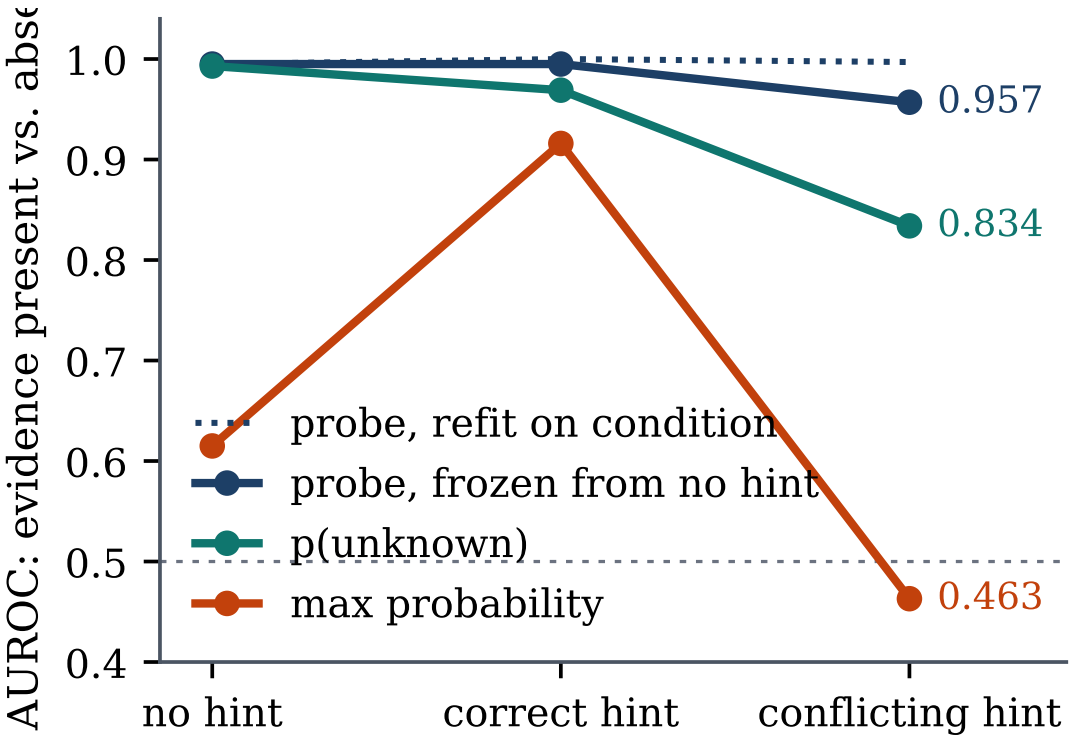


Figure 1: Evidence detection under three hint conditions (Qwen2.5-VL-3B). Native output signals degrade under conflict; the frozen final-state probe's ranking degrades much less, and its present–absent separation dilutes and rotates (Table 3). Values are AUROC over the 522 paired items; the dotted line is chance.

4.3 Dilution and rotation in the full space

The probe-score gap of the previous paragraph is a projection onto a single direction, the unit no-hint mean difference w. To separate length from direction we compute, for each condition, the full mean-difference vector $\Delta\mu = \bar{h}_{\text{present}} - \bar{h}_{\text{absent}}$ and report its norm relative to the no-hint norm, its angle to w (the principal angle between the two one-dimensional subspaces), its projection $\Delta\mu \cdot w$, its length after whitening by the no-hint within-class covariance (Mahalanobis distance; ridge 1% of the mean eigenvalue), and the coherence of the per-scene paired gaps $v_s = h_{\text{present},s} - h_{\text{absent},s}$, defined as $\|\bar{v}_s\|/\text{rms}_s\|v_s\|$. All quantities carry scene-grouped bootstrap intervals (500 draws). The same estimator applied to the no-hint vector against itself gives ratio 1.00 $[0.97, 1.05]$ and an angle of 3°–5°, which is the noise floor of the measurement.

Under a conflicting hint the full vector shortens to 0.53 $[0.50, 0.55]$ (hint `a`) and 0.57 $[0.54, 0.60]$ (hint `b`) of its no-hint length in 3B, to 0.39/0.43 in 7B and 0.61/0.64 in LLaVA, and turns 39°/41°, 48°/44° and 71°/69° away from w—eight to sixteen times the noise floor. The projection onto w falls to 0.41/0.43, 0.26/0.31 and 0.20/0.23, further than either factor alone, because $\|\Delta\mu\| \cos\theta$ is their product. Two different hinted colours produce almost the same direction: $\cos(\Delta\mu_{\texttt{a}}, \Delta\mu_{\texttt{b}}) = 0.908$, 0.908 and 0.915 in 3B, 7B and LLaVA. The two pairs share the present arm and differ only in the hinted colour and the absent arm, so this agreement means the shift is not specific to the colour being asserted: it is a common response to receiving a hint. A correct hint also shortens and rotates the vector (0.69–0.78 of its length, 22°–46°) but keeps a larger along-axis projection (0.61 against 0.41 in 3B; 0.69 against 0.26–0.31 in 7B; 0.48 against 0.20–0.23 in LLaVA), which is consistent with its undamaged ranking. In 3B the correct hint rotates as much as the conflicting one, so the angle alone does not predict damage; what separates the two hints is how much of the length survives.

The information is re-coded rather than deleted. Whitened by the no-hint within-class noise, the separation does not fall at all: Mahalanobis distance goes from 0.45 to 0.52/0.49 in 3B, 0.43 to 0.55/0.53 in 7B and 0.42 to 0.63/0.62 in LLaVA, and a readout refitted inside the condition reaches 0.995–0.999. What does fall is agreement across scenes. Per-scene gaps keep most of their length (78.5 → 61.7 in 3B, 155.4 → 102.9 in 7B, 108.4 → 87.9 in LLaVA) but stop pointing the same way: coherence drops from 0.60 to 0.40, from 0.70 to 0.41 and from 0.61 to 0.45. The aggregate vector is short for two reasons—each gap is shorter, and shorter gaps disagree in direction—and a fixed linear readout sees only the aggregate. This is the representational fact behind the output collapse: $p(\texttt{unknown})$ and maximum probability are absolute, scale-dependent readings of a representation that

Table 3: Geometry of the present–absent mean difference under hints (261 scenes; `a` plausible, `b` far hue; "Gemma" is Gemma-3-4B-it). "Ratio" is $\|\Delta\mu_{\text{cond}}\|/\|\Delta\mu_{\text{none}}\|$; "angle" is to the no-hint axis w; "along" is the projection on w relative to the no-hint projection. Mahalanobis and coherence show the no-hint value, then the hinted value; the Mahalanobis length is in units of the no-hint within-class noise, and coherence is $\|\bar{v}_s\|/\text{rms}_s\|v_s\|$.

model	hint	ratio	angle	along	Mahalanobis	coherence
3B	`a`	0.53	39°	0.41	0.45 → 0.52	0.60 → 0.40
	`b`	0.57	41°	0.43	0.45 → 0.49	0.60 → 0.44
7B	`a`	0.39	48°	0.26	0.43 → 0.55	0.70 → 0.41
	`b`	0.43	44°	0.31	0.43 → 0.53	0.70 → 0.47
LLaVA	`a`	0.61	71°	0.20	0.42 → 0.63	0.61 → 0.45
	`b`	0.64	69°	0.23	0.42 → 0.62	0.61 → 0.49
Gemma	`a`	0.53	29°	0.47	0.39 → 0.42	0.59 → 0.54
	`b`	0.54	24°	0.50	0.39 → 0.42	0.59 → 0.56

has been diluted and rotated, whereas a rank-based readout keeps most of its accuracy because the ordering survives. Neither quantity orders the models the way their behaviour does (7B dilutes most yet adopts hints least; LLaVA rotates most and adopts most), so the geometry is a shared consequence of the prompt rather than a readout of credulity. The measurement is descriptive: it establishes that a hint does not remove visual evidence from the state, and it bounds what a fixed direction can recover, but it does not identify the weights that apply the transformation.

Table 3 adds a backbone outside the Qwen family. In Gemma-3-4B-it a conflicting hint shortens the evidence vector to 0.529 [0.490, 0.567] (hint a) and 0.543 [0.502, 0.580] (hint b) of its no-hint length and turns it 29°/24° away from w, five to six times its own noise floor (1.00 [0.94, 1.06], 4.5°). The two hinted colours point in almost the same direction (cos = 0.971, higher than in any Qwen-family model), and a probe fitted on training scenes' no-hint states scores held-out scenes' hinted states at 0.964 [0.950, 0.976] (a), 0.970 [0.957, 0.981] (b) and 0.981 [0.971, 0.989] (correct), against refits of 0.989–0.991. What does not replicate is the loss of cross-scene coherence (0.59 → 0.54/0.56) and, most clearly, the collapse of the output signals: p(`unknown`)'s AUROC falls only from 0.961 to 0.891 and maximum probability *rises* from 0.627 to 0.701, although the answers are overridden as strongly as in 3B (accuracy 0.567 → 0.241, adoption 0.674). The internal geometry therefore generalises across language-model families, while the output-space collapse that motivates a frozen gate is a property of the Qwen2/2.5 readouts; in Gemma the gate's margin over p(`unknown`) is smaller (0.964 against 0.891) but still present. The decision-level decomposition is model-specific as well: on the fixed no-hint axis Gemma shows a criterion shift with no detectable sensitivity loss ($\Delta d' = -0.44$ [−0.88, +0.17], $\Delta c = +1.25$ [+0.72, +1.26], scene bootstrap), where 3B loses sensitivity (3.51 → 1.45). What survives in all four models is the geometry, not the signal-detection profile.

4.4 Dilution, not a criterion shift

A natural reading of Table 1 is that the hint lowers the refusal threshold. Signal detection separates this from a loss of sensitivity. Using the out-of-fold probe score as the evidence axis, we compute d' and criterion c for each condition (Appendix Table 14), and fit the psychometric model logit $P(\text{refuse}) = b_0 + az + \sum_k d_k(b_k + a_k z)$ over the 1,827 item–condition observations. The fit is run twice: on the *fixed* axis, z is the no-hint state of the same item; on the *own* axis, z is the condition's own state projected on the same no-hint direction.

Under conflict, sensitivity falls from 3.51 to 1.45 ($\Delta d' = -2.13$ [−2.69, −1.66]), while the criterion does not move reliably in this model ($\Delta c = +0.16$ [−0.18, +0.39]). A correct hint leaves both unchanged ($\Delta d' = +0.31$ [−0.36, +0.93]). The behavioural effect is bidirectional—conflict raises refusal on present items and lowers it on absent items—and a criterion shift moves both arms in the same direction, so the criterion account cannot produce the pattern.

On the fixed axis the conflict slope is −0.171 against −0.719 (change +0.548 [0.439, 0.774]; likelihood-ratio 178.2, $df = 2$, $p < 10^{-38}$), a 4.2-fold flattening. Most of that is the axis. The hint halves the separation on the probe's direction (Section 4.3 shows this to be dilution plus rotation), so a one-unit change on the fixed axis corresponds to about half a unit on the direction the decision reads. Applying the condition's own slope to the diluted axis predicts a 2.1-fold flattening; the remainder comes from imperfect coupling between the pre- and post-hint projections (correlation 0.86). When z is taken from the conflict state itself, the slope change is +0.160 [0.013, 0.429] (LR 6.4, $p = 0.041$): in its own coordinates the decision is 22% flatter. That residual is

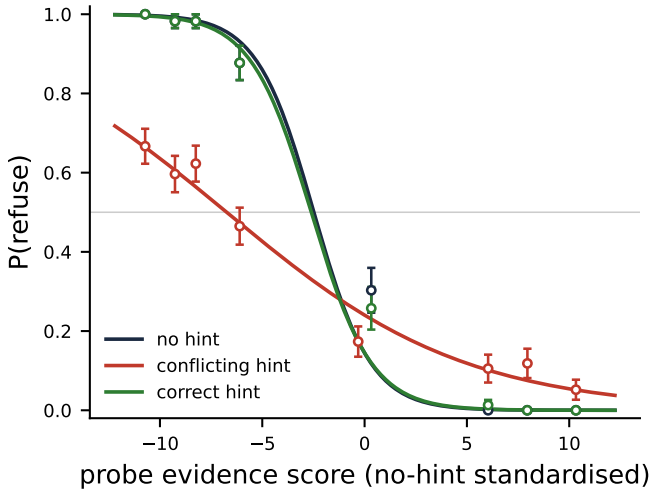


Figure 2: Refusal probability versus the no-hint probe score, measured before the hint is applied. Points are binned rates with binomial standard errors; lines are fitted psychometric curves. The conflicting-hint curve is flatter on this fixed axis and pivots around $z \approx -1.2$; most of the flattening is the dilution and rotation of the axis, with a 22% residual on the condition's own axis. The correct-hint curve tracks the no-hint curve.

small but nonzero, and we report it as a real sensitivity change rather than absorbed by the dilution of the axis.

The sensitivity collapse replicates in all three models; the criterion does not. Its shift is null in 3B, negative in 7B (-0.54 $[-0.73, -0.35]$) and positive in LLaVA ($+1.00$ $[+0.66, +1.33]$). A uniform lowering of the refusal threshold predicts a common direction, so the cross-model pattern argues against a criterion account. It also warns against over-reading c: the equal-variance estimate is biased when d' changes, which is why we treat the consistent d' loss, not the sign of Δc, as the invariant. Under a correct hint d' never falls ($\Delta d' = +0.31$, $+0.50$, $+0.32$ for 3B/7B/LLaVA), so the loss is specific to conflict. The flattening is 4.2-fold in 3B; we do not use the word "fourfold" for the other models, where the fixed-axis changes are $+0.332$ (LR 164.8) and $+0.286$ (LR 40.8). On each condition's own axis those fall to $+0.113$ $[0.025, 0.232]$ (LR 22.0) and $+0.053$ $[-0.057, 0.185]$ (LR 1.6, n.s.), residual gain losses of 24% and 12%.

4.5 Causal evidence: sufficiency at the readout, localisation late

Table 4 closes the gap between "decodable" and "used" at the readout. The direction is fitted before any hint and injected during the conflicting pass. In 3B, subtracting w raises absent-object refusal from 0.542 to 0.874 at $-0.25\,w$ ($+0.331$ $[0.289, 0.374]$) and to 0.941 at $-0.50\,w$, near the 0.904 no-hint baseline; on present items it raises refusal from 0.088 to 0.759. Adding w drives absent refusal to 0.065 at $+0.25\,w$ and to 0.000 at $+0.50\,w$. The effect is monotone and saturates. The held-out split-half fit reproduces it, and the other two models move in the same directions from base policies as different as 91.8% and 13.8% absent refusal. The same direction also opens the gate in LLaVA, which is already credulous, and closes it in 7B, which over-refuses. Sufficiency at the readout is therefore established in all three models.

The direction does not resolve the colour. In 3B, $+0.25\,w$ raises true-colour answers from 0.383 to 0.479 ($+0.096$ $[0.050, 0.142]$) while hint adoption stays at 0.433 (-0.015 $[-0.065, 0.034]$). In LLaVA, $+0.25\,w$ leaves true-colour answers statistically unchanged ($0.360 \to 0.379$, $+0.019$ $[-0.011, 0.050]$) while raising hint adoption on absent items from 0.833 to 0.990. The gate decides whether the model answers; the hint still wins the content. This is the causal analogue of the psychometric result: the axis remains connected to the gate, but it is diluted, rotated, and under-weighted.

Removal is not direction-specific. Sufficiency does not imply necessity, and the raw necessity test is biased. Projecting out w lowers absent refusal by 0.214 (3B), 0.211 (7B) and 0.102 (LLaVA), but w carries 5.4× more absolute projection energy than a random direction on average (mean fraction 0.055 against 0.019, 0.004 and 0.008; 5.3× in LLaVA), so the raw comparison pits directions that remove different amounts of the state. We therefore match the removed energy in both directions (Table 5). Scaling w down to the energy of a random direction reduces its effect to -0.073 $[-0.096, -0.052]$ in 3B and -0.018 $[-0.031, -0.008]$ in 7B. Scaling a random

Table 4: Steering and ablation during the conflicting-hint pass ($n = 261$ per model). The direction is fitted separately on each model's no-hint states and injected at the unembedding input; positive α pushes toward "evidence present". The two absent-arm columns average hints a (plausible) and b (far-hue); the base refusal rate is 0.521 and 0.467 hint adoption for hint a alone in 3B. "Rand-ablate" is the largest effect of projecting out any of three random directions at their own energy.

model	setting	present refusal	present true	absent refusal	absent hint
3B	base	0.088	0.383	0.542	0.450
	$-0.25\,w$	0.759	0.073	0.874	0.125
	$+0.25\,w$	0.000	0.479	0.065	0.845
	ablate w	0.031	0.429	0.328	0.651
	rand-ablate (max)	0.091	0.391	0.548	0.461
7B	base	0.556	0.249	0.918	0.082
	$-0.25\,w$	0.950	0.011	1.000	0.000
	$+0.25\,w$	0.004	0.421	0.057	0.870
	ablate w	0.368	0.345	0.707	0.291
	rand-ablate (max)	0.566	0.253	0.921	0.088
LLaVA	base	0.004	0.360	0.138	0.833
	$-0.25\,w$	0.345	0.149	0.734	0.241
	$+0.25\,w$	0.000	0.379	0.000	0.990
	ablate w	0.000	0.356	0.036	0.946
	rand-ablate (max)	0.004	0.364	0.145	0.827

Table 5: Energy-matched removal during the conflicting-hint pass ($n = 261$). "Base" is refusal on absent items, averaged over hints a and b. Each ablation removes a direction rescaled so that its mean removed energy matches the stated target: $w{\to}r$ scales w to a random direction's energy, $r{\to}w$ scales a random direction to w's energy, and $cm{\to}w$, $pc1{\to}w$ do the same for the class-mean and first-PC directions. Brackets are 95% scene-bootstrap intervals on the change from base. The raw w column removes 5.4× more energy than a random direction.

model	base	w (raw)	$w{\to}r$	$r{\to}w$	$cm{\to}w$	$pc1{\to}w$
3B	0.542	0.328	0.469	0.552	0.625	0.625
		$[-.253, -.180]$	$[-.096, -.052]$	$[-.004, .025]$	$[.052, .111]$	$[.052, .111]$
7B	0.918	0.707	0.900	0.874	0.921	0.921
		$[-.247, -.178]$	$[-.031, -.008]$	$[-.063, -.027]$	$[-.011, .019]$	$[-.011, .019]$
LLaVA	0.138	0.036	0.126	0.109	0.075	0.100
		$[-.130, -.075]$	$[-.021, -.004]$	$[-.046, -.013]$	$[-.088, -.042]$	$[-.057, -.021]$

direction up to the energy of w is null in 3B ($+0.010$ $[-0.004, +0.025]$) but not in 7B (-0.044 $[-0.063, -0.027]$), where it is larger than the matched effect of w. The class-mean and first-PC directions, scaled to the same energy, leave refusal unchanged in 7B ($+0.004$ $[-0.011, +0.019]$) and raise it in 3B ($+0.083$ $[+0.052, +0.111]$). In LLaVA the gate is already near the floor (base refusal 0.138), and there the matched effect of w (-0.011 $[-0.021, -0.004]$) is smaller than that of a matched random direction (-0.029 $[-0.046, -0.013]$) and of the matched class-mean direction (-0.063 $[-0.088, -0.042]$). No direction is necessary in a direction-specific sense: most of the raw removal effect is removed energy, and the residual effect of w is no larger than that of a matched random direction. What survives is sufficiency. Injecting w at the readout moves the gate monotonically, including when the dose is scaled to each prompt's own final-state norm (absent refusal $1.000 \to 0.134$ in 3B and $1.000 \to 0.073$ in 7B between -0.25 and $+0.25$), so the dose–response is not an artifact of a fixed absolute norm. We describe w as an efficient readout direction for evidence availability, not as an identified circuit element.

Localisation. Patching answers where restoring the unpolluted state reopens the gate. In 3B, patching at layer 24 (67% depth) recovers 22% of the lost refusal and $p(\texttt{unknown})$, and layer 32 (89%) recovers 80%. In 7B the corresponding values are 0%, 69% and 90% at layers 18, 24 and 26 (64, 86, 93%); in LLaVA they are 36%, 80% and 96%. Recovery is concentrated in the last few layers. A shuffled-source control recovers as much or more (3B L32 0.927; 7B L26 0.965; LLaVA L26 0.842), so the restored state carries a category-level "no evidence" status, not a sample-specific code, and the comparison cannot separate evidence from a generic

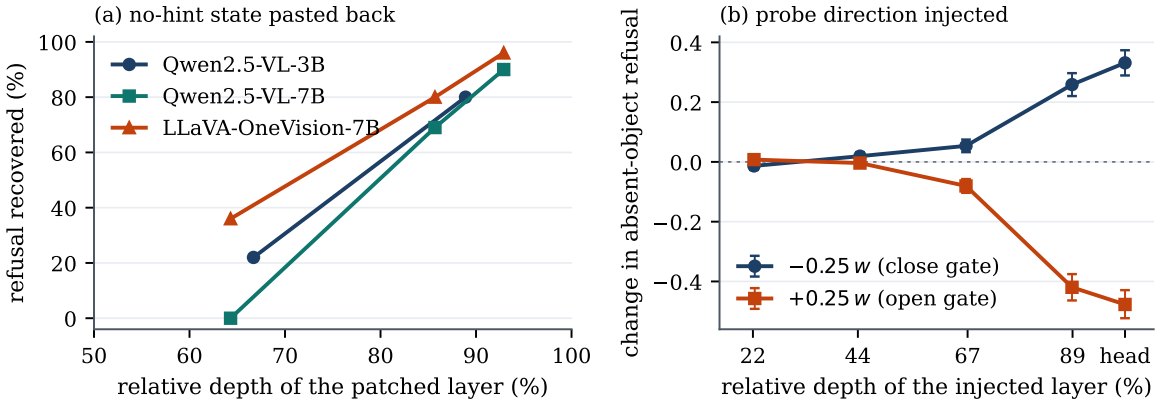


Figure 3: Localisation by restoration and by injection. (a) Recovery of the refusal lost under conflict when the no-hint state is pasted back at layer L, in three models; depth is relative to decoder depth. (b) Change in absent-object refusal when the probe direction is injected at layer L of the 3B model with the dose matched to each layer's median norm; "head" is the unembedding input. Both profiles are late and steep.

no-hint prior. A same-image control does separate the question: with the image fixed, patching the image's own absent-question state at L32 raises refusal from 0.521 to 0.839, whereas its present-question state drives refusal to 0.008 (paired difference +0.83 [0.79, 0.87]), with the same contrast in 7B (+0.84) and LLaVA (+0.74). The late state therefore encodes a question-conditioned answer/refuse policy. A no-op patch that pastes the same pass's own state returns the base distribution exactly (argmax agreement 1.00 and maximum probability change 0 at both probed layers in 3B and 7B, and in LLaVA), whereas a zero paste changes the argmax in 10.7%/17.2% of absent items in 3B, 38.7%/6.1% in 7B and 72.4%/99.6% in LLaVA. The hook is therefore active and the pasted vector, not the instrumentation, drives the change, so the recovery profile can be read quantitatively; the caveat is that pasting a foreign state still shifts the input distribution. The claim is that the gate is written late, not that we have identified the computation.

LLaVA-OneVision-7B provides the strongest available generality test. It uses a different vision encoder and training recipe but shares the Qwen language backbone. Its present-item refusal under conflict is 0.004, against 0.556 in Qwen-7B: the same hint is much more effective on this model. Yet the recovery profile has the same shape and depth, so a model that is easier to fool does not have an earlier gate; its gate is pushed further open at the same place.

4.6 A frozen gate, and where it stops helping

Table 6 evaluates the gate. At 50% coverage under conflict, ranking by the frozen probe reduces fabrication from 0.249 to 0.069 in 3B (3.6×), from 0.268 to 0.126 in 7B (2.1×), and from 0.295 to 0.111 in LLaVA (2.7×); maximum probability is far worse in every model. The refit probe reaches 0.011 in 3B, but that number uses a decoder fitted on the same condition and is not deployable; we report it only to show that the frozen score is not at the decodability ceiling. A logistic gate over the 12 output probabilities reaches 0.142 at the same coverage, so the gain is not merely a learned readout of the output distribution.

The advantage survives prevalence shift: applying the 3B threshold to unanswerable prevalences of 5, 10, 25 and 50% yields fabrication among answered of 0.007, 0.016, 0.045 and 0.125, against 0.032, 0.066, 0.174 and 0.387 for $p(\texttt{unknown})$, with the same ordering in 7B and LLaVA. Conformal risk control gives a finite-sample certificate, but a loose one at this sample size. With 130 calibration scenes a 5% target is infeasible for 3B and LLaVA (minimum upper bound 0.061 and 0.059), and a 10% target is certified at coverage 0.519 and 0.511. In 7B the 5% certificate does not transfer: it yields 0.067 on the held-out half, above target, and we report this rather than selecting the favourable split. A pollution detector built as probe $- p(\texttt{unknown})$ is not useful: its AUROC is 0.97/0.95/0.96, but at 50% coverage it selects the same items as the probe alone.

4.7 External validation and task form

On VizWiz [10] (462 items; Appendix Table 11) conflict lowers unanswerable refusal from 0.704 to 0.636 and raises answerable refusal from 0.099 to 0.193 while dropping accuracy from 0.580 to 0.451. The frozen probe transfers at 0.827/0.720, above maximum probability but below $p(\texttt{unknown})$ (0.878/0.793): natural unanswerable

Table 6: Deployment under conflicting hints (hint `a`), out-of-fold and frozen scores. Panel A: fabrication at 50% coverage. Panel B: fabrication among answered items after applying the 3B 50%-coverage threshold to unanswerable prevalences of 5–50% (probe rows; p(`unknown`) in parentheses). Panel C: split conformal risk control with 130 calibration scenes; "—" marks infeasible levels.

A: fabrication at 50% coverage				
model	probe	p(`unknown`)	max prob.	
3B	**0.069**	0.249	0.510	
7B	**0.126**	0.268	0.648	
LLaVA	**0.111**	0.295	0.655	
B: fabrication among answered, prevalence sweep				
model	5%	10%	25%	50%
3B	0.007 (0.032)	0.016 (0.066)	0.045 (0.174)	0.125 (0.387)
7B	0.028 (0.050)	0.058 (0.100)	0.155 (0.250)	0.355 (0.500)
LLaVA	0.007 (0.023)	0.014 (0.048)	0.042 (0.132)	0.116 (0.314)
C: conformal risk control (probe)				
model	$\alpha = 5\%$	$\alpha = 10\%$	coverage	test fab.
3B	infeasible (UCB 0.061)	certified	0.519	0.088
7B	certified	—	0.340	0.067
LLaVA	infeasible (UCB 0.059)	certified	0.511	0.127

questions fail for heterogeneous reasons, and the hint does not attack the refusal gate. On VQA-v2 colour questions the ordering survives but calibration does not: the score shifts from +6.93 to +2.10 ($r = 0.54$), and maximum probability has lower error at 50% coverage (0.007 vs. 0.087). A task-form swap (Appendix Table 15) leaves the gate intact but collapses the output readout mainly for colour (p(`unknown`) loses 0.159, vs. 0.023 and 0.040), and the gate's advantage tracks that damage (3.6× vs. about 2×).

5 Discussion

What the hint does. The hint does not erase the evidence; it dilutes and rotates it, and changes a late decision about whether to report it. A fresh readout on the hinted state still ranks the arms at 0.989–0.999, but the no-hint direction loses more than half its separation, and that loss decomposes into a shorter mean difference (0.39–0.64 of its length) and a systematic turn away from the direction (24°–71°), with the per-scene gaps also falling out of agreement in the Qwen-family models. A direction added at the readout moves the decision. The most parsimonious account is a readout that reads a fixed direction, fed by an evidence-availability representation computed upstream whose content is preserved but re-coded by the assertion, with the abstention decision written late. It is not yet a mechanism: the steering is at the readout, energy-matched removal shows that no single direction is necessary, and the geometry analysis bounds what a fixed direction can recover without identifying the weights that apply the transformation.

When the gate helps. In the constructed regime the hint attacks the refusal gate and p(`unknown`) is corrupted; there the frozen probe buys a factor of two to four. On naturally unanswerable images the failure is heterogeneous and p(`unknown`) is already good, so the probe adds little. A practical system should use the internal gate when it suspects prompt pollution, operate it by coverage rather than a fixed threshold, and fall back to calibrated confidence when evidence is present.

Limitations. The benchmark is small (261 images) but densely observed. Absent categories are sampled without requiring scene plausibility, and "absent" is annotation-level; a co-occurrence score for the absent category against the image's own objects predicts no-hint refusal (Appendix D), so the gate partly tracks scene fit, though evidence detection survives within both strata (p(`unknown`) AUROC ≥ 0.976) and the effect survives a control for category frequency. On the main absent arm a yes/no presence probe gives mean $p(\text{yes}) = 0.046$ and judges 2.7% of items present above 0.5; restricting to the 254 items it judges absent leaves refusal under conflict at 0.531 (against 0.521 on all items), hint adoption at 0.457, and probe AUROC at 0.999. LLaVA's own presence readout is weak (mean $p(\text{yes}) = 0.403$), so for LLaVA absence rests on the annotation, though

its strict subset gives the same picture (refusal 0.104, adoption 0.891, probe 1.000). The main prompts offer an explicit `unknown` option and a bracketed hint. Removing the `unknown` option removes abstention by construction (refusal 0) and raises hint adoption from 0.467 to 0.935 in 3B and 0.123 to 0.977 in 7B; phrasing the hint as a sentence rather than a bracket restores refusal to 0.958/0.989 and all but eliminates hint adoption (0.038/0.004), while accuracy under the wrong hint rises from 0.383 to 0.594 in 3B and 0.249 to 0.498 in 7B. LLaVA moves the same way from a lower baseline: sentence phrasing raises conflict refusal from 0.088 to 0.559, cuts adoption from 0.904 to 0.333 and raises wrong-hint accuracy from 0.360 to 0.548. The conflict effect is therefore specific to a prefixed assertion with an available abstention token; we do not claim it for conversational hints or for readouts without `unknown`. The mechanistic claims are strongest for colour. The probe requires white-box states. Energy-matched removal shows no direction-specific necessity, so we claim readout-level sufficiency, not a circuit. To break the confound between architecture and training recipe we added Gemma-3-4B-it (Gemma-3 with a SigLIP encoder), which abstains without hints (0.843 of absent items) and is overridden by conflict (0.402/0.475 refusal, adoption 0.579/0.502). Its evidence vector dilutes to 0.53 and rotates 24°–29°, its two hinted colours coincide at cos = 0.971, and its frozen probe transfers at 0.964/0.970, so the geometry replicates outside the Qwen family; its output confidence, however, barely degrades (p(`unknown`) AUROC 0.961 → 0.891, maximum probability 0.627 → 0.701). The rank–scale dissociation is therefore about the representation; the collapse of the native output signal is a property of the Qwen2/2.5 readouts, and we scope it accordingly. SmolVLM-Instruct (Idefics3 with a SmolLM2 language model) remains a boundary case with no abstention behaviour at all (refusal 0 in every cell), so it cannot test the gate. The geometry analysis is descriptive: it establishes dilution, rotation and loss of cross-scene coherence, and bounds what a fixed direction recovers, but a rotated yet fully decodable axis is compatible with several implementations and we do not identify which. Input resolution is not the driver, since moving the 3B model from its native 589,824 to 262,144 pixels changes absent refusal only from 0.521 to 0.556 and leaves the frozen probe at AUROC 0.997. The conformal certificate is marginal at 130 calibration scenes. The invariance is rank invariance, not calibration invariance. Labels were verified by a single annotator on contact sheets; we did not run a second-annotator agreement study.

6 Conclusion

A conflicting hint does not make a vision-language model blind to the image. It dilutes and rotates the internal evidence axis—shortening the present–absent mean difference to 0.39–0.64 of its no-hint length and turning it 24°–71° off the direction a no-hint readout uses—and overrides a late decision to abstain. The classes remain decodable (a refit recovers at least 0.989) and the ranking survives on held-out scenes; the scale and the alignment do not, which is why in the Qwen-family models the model's own confidence collapses while a freshly fitted readout does not. The override is localised to the last few layers in the three white-box models; the behavioural override and its internal geometry replicate across four models from two language-model families (Qwen and Gemma) and two vision encoders; and the direction is bidirectionally controllable at the readout where we tested it, while energy-matched removal shows that it is not necessary. The practical consequence is a gate fitted without model training that cuts fabrication by a factor of two to four at moderate coverage, with the boundary that on naturally unanswerable images p(`unknown`) is already as good. The scientific question is what applies the dilution and the rotation, and whether a gate can be trained to ignore assertions that the image contradicts.

Reproducibility statement

The repository at https://github.com/yuanchongbit/evidence-gate contains the frozen manifest with verified labels, all runners, probes, steering and patching code, and the commands for every table. Raw logs are under `results/runs/`; all experiments use public data and one 24 GB GPU.

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A Prompts, readout and regularisation

System prompt:

```
You are a careful visual question answering assistant.  Answer with exactly one word.  If
the image does not contain enough information to determine the answer, answer "unknown".
```

User prompt for the colour task:

```
[Hint:  the {category} is {colour}.]
What color is the {category}?
Answer with one word.
```

The side task asks "Is the {category} on the left or on the right? Answer with one word."; the count task asks "How many {category} are in this image? Answer with one word." with candidates `one`, `two`, `three`, `unknown`. All twelve colour candidates are single tokens under each tokenizer's leading-space convention.

The AUROC probe uses standardised features and $C = 0.1$ with scene-grouped five-fold cross-validation. The steering direction uses $C = 0.01$ on the same pair, because it is injected at a fixed norm and a less regularised direction is dominated by high-variance dimensions; the two settings differ only in shrinkage and are stated wherever a number depends on them. The 12-dimensional logit readout is logistic regression on the twelve candidate probabilities with $C = 1.0$.

B Scene-held-out transfer and regularisation sensitivity

The headline frozen numbers fit the probe on the no-hint states of four scene folds and score the fifth fold's hinted states, so no scene contributes to both fitting and evaluation. The protocol matters: scoring the probe on the same scenes it was fitted on gives 0.971 for the far-hue hint in 3B against 0.958 $[0.939, 0.974]$ held out, and the inflation is 0.002–0.035 across the four models and three hint conditions (largest for LLaVA, $0.958 \to 0.923$; Gemma-3-4B inflates by 0.010–0.019, $0.982 \to 0.964$ for the plausible hint). Scene overlap therefore helped the earlier numbers estimate, but it does not account for the ranking: the held-out probe still separates the conflict arms far above chance and within 0.04 of its fitting-scene value.

The regularisation is fixed at $C = 0.1$ for every reported probe number. Sweeping $C \in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 0.1\}$ under the held-out protocol moves the conflict AUROC by at most 0.020 (LLaVA, plausible hint, 0.906–0.926), 0.012 in 3B and 0.016 in 7B; for the far-hue hint the spread is at most 0.016 in every model. No choice of C changes the ordering—conflict below no hint, correct at or near ceiling—or the size of the output-signal collapse, which does not depend on the probe at all. We report the fixed $C = 0.1$ rather than the grid-selected value so that the headline number is not chosen on the evaluation set.

C Per-colour breakdown

Table 7: Verified samples by gold colour. Counts are the 261 manifest items; the absent arm uses the category's most frequent colour, so it is not restricted to this set.

colour	blue	white	black	green	red	yellow
n	88	79	45	25	15	9

Blue and white dominate the manifest. A per-colour AUROC is not estimable at useful precision for red and yellow ($n = 15$ and 9), and we do not report one. The colour distribution is a property of COCO val2017 under our geometric filter, not a design choice; the absent hint uses each category's own colour distribution.

D Controls for text priors and strict absence

A gray image with the same prompts and dimensions yields probe AUROC 0.849/0.981/0.930 under no/conflicting/correct hints, and p(`unknown`) 0.700/0.325/0.433. Text alone carries some category-to-colour prior, enough to reach 0.85 on the no-hint pair. But the direction learned from gray images transfers across hints at only 0.540/0.603, whereas the real-image direction transfers at 0.958/0.990 on held-out scenes (Table 2). The real-image direction is shared across prompts; the text-prior direction is not.

Category-matched pairs hold the question fixed and pair each present image with another image in which the category is annotated absent. Cross-validation groups by category, so a held-out fold contains categories never seen during fitting. The probe reaches 0.985 with no hint and 0.973 under conflict; p(`unknown`) drops from 0.983 to 0.817 and maximum probability from 0.701 to 0.468. Excluding the 17 of 260 negatives that the presence probe judges present above 0.5 leaves this unchanged: p(`unknown`) is 0.992 without hints and 0.851 under conflict, and maximum probability 0.692 and 0.473. The main absent arm is checked the same way: per-item presence probabilities on all 261 absent images give mean $p(\text{yes}) = 0.046$, and 7 items are judged present above 0.5. On the remaining 254, refusal under the conflicting hint is 0.531 (against 0.521 on all items), refusal with no hint is 0.921, hint adoption is 0.457, and the evidence probe reaches AUROC 0.999; the strict-absence

restriction therefore does not change the main result. For LLaVA the presence check is much weaker (mean $p(\text{yes}) = 0.403$, 73 items above 0.5), so its yes/no readout is not a reliable absence detector; on the 183 items it judges absent, conflict refusal moves only from 0.088 to 0.104, adoption is 0.891 and the probe remains at 1.000. For LLaVA, "absent" therefore rests on the annotation rather than on the probe.

A model without an abstention behaviour. We also ran the main battery on SmolVLM-Instruct (2B; Idefics3 vision encoder, SmolLM2 language model) as a non-Qwen backbone. It never emits `unknown`: refusal is 0 in all six cells, and it adopts the stated colour on 0.985 (hint a) and 0.989 (hint b) of absent items, so there is no behavioural gate to study and we do not use it as a replication. Its final state nonetheless separates present from absent items, at probe AUROC 0.989 with no hint and 0.993 under the conflicting hint (p(known) 0.921 and 0.598; maximum probability 0.877 and 0.257). A decodable evidence axis can therefore exist in a model that never abstains. Because the conflict-pair probe for this model could also pick up hint–image mismatch rather than evidence, we treat this as a boundary observation about the axis, not as evidence about the gate.

Scene fit of the absent category. The absent category is sampled without regard to whether it fits the scene, so abstention could track implausibility rather than missing evidence. For each item we score the absent category by its COCO train co-occurrence with the categories annotated in that image, $s = \text{mean}_c\, P(\text{absent} \mid c)/P(\text{absent})$, and split at the median (Table 8). The score predicts no-hint refusal in every model, so the absent arm does carry a scene-fit component: the better-fitting half refuses less by 0.116 $[0.047, 0.185]$ in 3B, 0.069 $[0.023, 0.115]$ in 7B and 0.132 $[0.039, 0.224]$ in LLaVA. Category frequency is a rival explanation, since frequent categories are also refused less in 7B (0.073 $[0.025, 0.128]$) and LLaVA (0.100 $[0.008, 0.195]$), though not detectably in 3B (0.062 $[-0.014, 0.133]$). The two scores are only weakly related (Spearman $\rho = 0.23$). Fitting no-hint refusal on both jointly with a scene bootstrap, scene fit remains negative in all three models, at -0.81 $[-1.26, -0.37]$ (3B), -0.68 $[-1.24, -0.29]$ (7B) and -0.44 $[-0.78, -0.16]$ (LLaVA) per standard deviation, while frequency is negative only in 7B (-0.45 $[-0.98, -0.14]$) and indistinguishable from zero in 3B (-0.29 $[-0.76, +0.11]$) and LLaVA (-0.01 $[-0.34, +0.37]$). The scene-fit difference is also present within each frequency half (Table 9). It does not account for the evidence signal. Within each half p(`unknown`) still separates present from absent images, at AUROC 0.988–0.998 in 3B, 0.976–0.991 in 7B and 0.986–0.995 in LLaVA, and the probe at 0.992–0.999, 0.989–1.000 and 0.993–0.997. Under the conflicting hint the stratum difference is not significant in 7B (-0.047 $[-0.124, +0.037]$) or LLaVA ($+0.024$ $[-0.045, +0.093]$) and remains in 3B (-0.165 $[-0.287, -0.050]$), where hint adoption is 0.157 $[0.042, 0.280]$ higher in the better-fitting half. The same conclusion therefore holds in both strata: plausibility moves the level of the gate, not the ranking of evidence.

Table 8: Scene-fit stratification of the absent arm ($n = 261$ per model, median split). "lo"/"hi" are the lower and upper halves of $s = \text{mean}_c\, P(\text{absent} \mid c)/P(\text{absent})$, computed from COCO train co-occurrence with the categories annotated in the image. "AUROC" is p(`unknown`) separating present from absent under no hint; "probe" is the internal probe on the same pairs. Refusal and adoption use hint a.

model	half	n	refuse, none	refuse, a	adopt, a	AUROC / probe
3B	lo	131	0.962	0.603	0.389	0.998 / 0.999
	hi	130	0.846	0.438	0.546	0.988 / 0.992
7B	lo	131	0.992	0.901	0.099	0.991 / 1.000
	hi	130	0.923	0.854	0.146	0.976 / 0.989
LLaVA	lo	131	0.885	0.076	0.916	0.995 / 0.997
	hi	130	0.754	0.100	0.892	0.986 / 0.993

Table 9: No-hint refusal in the 2×2 of scene fit s (rows) and category frequency (columns), both median-split; cell sizes are 80, 51, 56 and 74 items from top to bottom. Scene fit lowers refusal in every frequency stratum.

cell	3B	7B	LLaVA
s lo, freq lo	0.963	1.000	0.887
s lo, freq hi	0.961	0.980	0.882
s hi, freq lo	0.893	0.982	0.839
s hi, freq hi	0.811	0.878	0.689

Prompt format. The main battery offers an explicit `unknown` option and a bracketed hint, which could overstate abstention or hint uptake. We rerun the absent arm three ways in all three models (Table 10). Removing the `unknown` option makes abstention impossible by construction (refusal is 0 in every condition) and raises adoption of the plausible hint from 0.467 to 0.935 in 3B, from 0.123 to 0.977 in 7B and from 0.904 to 0.992 in LLaVA. Phrasing the hint as a sentence rather than a bracket leaves refusal under conflict at 0.958/0.989 in 3B/7B, and raises it from 0.088 to 0.559 in LLaVA, while reducing hint adoption to 0.038/0.004/0.333; accuracy under the wrong hint rises from 0.383 to 0.594 in 3B, from 0.249 to 0.498 in 7B and from 0.360 to 0.548 in LLaVA. The conflict effect is thus tied to a prefixed assertion and to the availability of an abstention token; a conversational hint does not reproduce it.

Table 10: Prompt-format robustness ($n = 261$ absent items per model). "std" is the main battery; "no `unknown`" removes the `unknown` option from the system prompt; "sentence" phrases the hint as a sentence. Hint adoption for absent b is measured against the far-hue colour actually stated.

model	format	refuse, none	refuse, hint a	refuse, hint b	adopt a	adopt b
3B	std	0.904	0.521	0.563	0.467	0.433
	no `unknown`	0.000	0.000	0.000	0.935	0.908
	sentence	0.931	0.958	0.981	0.038	0.011
7B	std	0.958	0.877	0.958	0.123	0.042
	no `unknown`	0.000	0.000	0.000	0.977	0.973
	sentence	0.958	0.989	0.992	0.004	0.000
LLaVA	std	0.820	0.088	0.188	0.904	0.762
	no `unknown`	0.000	0.000	0.000	0.992	0.946
	sentence	0.793	0.559	0.736	0.333	0.088

Input resolution. LLaVA runs at a different input resolution from the Qwen models, so we rerun the 3B battery at 262,144 and 589,824 pixels. At 262,144, absent refusal is 0.908 with no hint and 0.556 under the plausible hint, hint adoption is 0.441, accuracy under the wrong hint is 0.433, and the frozen probe reaches AUROC 0.997; at 589,824 the values are 0.904, 0.521, 0.467, 0.383 and 1.000, matching the native run. Resolution moves the level by at most 0.05 and does not produce the conflict effect.

E External validation details

VizWiz is used for naturally unanswerable questions and VQA-v2 for colour questions with real answer distributions. The VizWiz run keeps 462 items (212 answerable, 250 unanswerable) and the same three prompt conditions; the hint is the most frequent colour of the queried category, as in the plausible condition. The VQA-v2 run uses 300 colour questions over 277 images, with the conflicting hint taken from the far-hue mapping. Both runs use the same 3B model and the probe frozen after fitting on the main no-hint pair.

Table 11: VizWiz (Qwen2.5-VL-3B). Refusal and accuracy by condition, and AUROC over answerable versus unanswerable questions. In-domain means refitting the probe inside VizWiz; frozen means transferring the probe fitted on the 261-image set.

condition	ans. refuse	ans. acc	unans. refuse	probe (in-dom.)	probe (frozen)	p(`unknown`)
no hint	0.099	0.580	0.704	0.909	0.827	0.878
correct hint	0.033	0.816	—	—	—	—
conflicting hint	0.193	0.451	0.636	0.803	0.720	0.793

On VizWiz the asymmetry is present but weaker than in the constructed design: the hint lowers unanswerable refusal by 0.068, whereas the constructed conflict lowers it by about 0.36, and it raises answerable refusal by 0.094. Maximum probability is close to chance on both pairs (0.437/0.367), while p(`unknown`) is strong (0.878/0.793). The frozen probe adds nothing over p(`unknown`) on this data (-0.051 $[-0.078, -0.023]$ without a hint, -0.074 $[-0.119, -0.031]$ with one). Naturally unanswerable questions are failed for heterogeneous reasons, and the hint does not target the refusal gate, so the regime in which the probe wins is not reached. At 50% coverage, the risk of an incorrect answer is 0.457 for the probe, 0.449 for p(`unknown`) and 0.655 for maximum probability; none of the three is usable, because half of these questions have no valid answer.

Table 12: VQA-v2 colour questions (300 items, 277 images, Qwen2.5-VL-3B). "Score" is the frozen probe score; "> 0" is the fraction it classifies as evidence-present. AUROC is for answering correctly on answerable questions. Risk is $1 -$ accuracy at the stated coverage.

condition	accuracy	p(`unknown`)	max prob.	score	> 0	AUROC
no hint	0.850	0.024	0.865	+6.93	0.940	0.710
correct hint	0.983	0.021	0.971	+6.70	0.983	—
conflicting hint	0.613	0.115	0.722	+2.10	0.780	0.489

VQA-v2 reproduces the direction of the effect without the constructed absent arm: the score falls by 4.83 points and the present-classification rate falls from 0.940 to 0.780, while accuracy falls from 0.850 to 0.613. The score is correlated across conditions ($r = 0.54$), so the shift is graded by the item rather than uniform. The ordering survives in the sense that the probe still separates correct from incorrect answers without a hint (0.710), but under conflict it is at chance (0.489), and maximum probability is the better correctness predictor in both conditions (0.831/0.649). At 50% coverage on no-hint answers, maximum probability has risk 0.007 against 0.087 for the probe. The probe is an evidence detector, not a correctness predictor, and on real answer distributions the two come apart.

F Per-layer probe

A probe refitted independently at each layer (grouped by scene) reaches AUROC 0.500 at the embedding output, 0.867 at layer 4, 0.982 at layer 8, and ≥ 0.990 from layer 16 onward; transfer from the no-hint pair to the conflicting pair has a dip at layer 8 (0.803) and recovers to 0.965–0.995 deeper. Refitting, however, gives each layer its own estimator, so this curve measures decodability rather than a shared axis. We therefore also fit the estimator once on the final layer (per fold) and apply its weights unchanged to every layer, with per-layer standardisation only. That shared readout is at chance through position 18 of the stored stack (embedding plus 36 layers; AUROC ≤ 0.53, rising to 0.58 and 0.64 at positions 19 and 21), then jumps to 0.86 at position 22, 0.97 at position 24 and 0.99 from position 25 onward. The rise is not monotone (0.67 at position 23), so we locate it to the last quarter rather than a single layer. Evidence is thus linearly decodable early, but in a layer-specific basis; the final-layer evidence axis used by steering and patching is assembled late. The two analyses share a target but not an estimator, and we do not map probe AUROC at layer L one-to-one onto recovery at layer L.

G Full dose–response

Table 13: Steering the no-hint direction during the conflicting pass (Qwen2.5-VL-3B, $n = 261$). "True" and "hint" are argmax rates on present items; "hint" on absent items is the rate of naming the hinted colour, averaged over hints a and b.

setting	present refusal	present true	absent refusal	absent hint
no hint, unsteered	0.011	0.762	0.904	—
conflict, base	0.088	0.383	0.542	0.450
$-0.50\,w$	0.931	0.008	0.941	0.057
$-0.25\,w$	0.759	0.073	0.874	0.125
$+0.25\,w$	0.000	0.479	0.065	0.845
$+0.50\,w$	0.000	0.479	0.000	0.665

H SDT across models and task forms

Criterion shifts under conflict are +0.16 [−0.18, +0.39] (3B), −0.54 [−0.73, −0.35] (7B) and +1.00 [+0.66, +1.33] (LLaVA); d' falls in all three. Under a correct hint, d' changes are +0.31, +0.50 and +0.32.

Table 14: Signal detection (Qwen2.5-VL-3B). d' and c are computed from paired present/absent refusal rates. Separation is the fixed probe's mean gap in pooled within-arm SD units. Slopes use z from the no-hint state (fixed) or the condition's own state (own), with the same direction in both.

condition	d'	c	separation	slope, fixed	slope, own
no hint	3.51	0.46	5.06	−0.719	−0.719
conflicting hint	1.45	0.62	2.60	−0.171	−0.559
correct hint	3.83	0.62	4.17	−0.680	−0.641

Table 15: Task form with the same images and absent categories (Qwen2.5-VL-3B). The gate opens in all three; the output readout collapses mainly for colour.

task	condition	refusal	adopts hint	probe	p(`unknown`)
colour	present, conflict	0.088	0.448	0.997	0.834
	absent, no hint	0.904	—	0.995	0.993
	absent, hint	0.521	0.467	—	—
side	present, conflict	0.000	0.268	0.991	0.966
	absent, no hint	0.621	—	0.993	0.989
	absent, hint	0.111	0.743	—	—
count	present, conflict	0.870	0.115	0.995	0.946
	absent, no hint	1.000	—	0.996	0.986
	absent, hint	0.996	0.004	—	—

I Deployment details

Scores are out-of-fold for the probe on the 3B model and frozen for the 7B and LLaVA models; all deployment numbers use the plausible hint `a` and pair `sufficient|wrong` with `absent|a`, giving 522 items. Fabrication is the fraction of answered items that are absent-object items; UAR is the same quantity divided by all absent items at full coverage. Conformal calibration uses half the scenes, a Clopper–Pearson upper bound at 90% confidence over all prefixes, and evaluation on the other half. The 3B refit gate reaches 0.011 at 50% coverage; the frozen score used in deployment reaches 0.069. The 12-dimensional logit readout reaches 0.142.

J Compute

All runs used one NVIDIA A10 with 24 GB. The main 3B run (six conditions per image, hidden states saved) takes about 30 minutes; the 7B and LLaVA runs take about an hour each; the Gemma-3-4B run takes about six minutes at a single 512-pixel resize; the full suite, including steering, patching and external validation, fits in a few GPU-hours.